

2.3.4 Hamiltonian Flow Maps are Symplectic

As we saw at the end of the last chapter, the variational equations describe the change in an infinitesimal perturbation of a solution of a dynamical system. For the Hamiltonian system $\dot{z} = J\nabla H(z)$, these take the form:

$$\dot{W} = JS(t)W,$$

where $S(t) = H_{zz}(z(t, \xi))$ is a symmetric matrix. Computing

$$W^T J \dot{W} = W^T J^2 S W = -W^T S W,$$

whereas

$$\dot{W}^T J W = W^T S^T J^T J W = W^T S W,$$

hence

$$\frac{d}{dt} W^T J W = W^T J \dot{W} + \dot{W}^T J W = 0.$$

This means that $W^T J W$ is a constant matrix. Observe that $W(t) = \mathcal{F}'_t(z(t, \xi))$ and that $W(0) = \mathcal{F}'_t(z(t, \xi))|_{t=0} = I$, hence

$$W^T J W \equiv W(0)^T J W(0) = J.$$

This proves that the flow map of a Hamiltonian system is a symplectic map.

2.3.5 The Symplectic Maps Form a Group

Let Φ_1 and Φ_2 be any pair of symplectic maps. Then

$$(\Phi_1 \circ \Phi_2)' = \Phi_1' \Phi_2',$$

by the chain rule, hence

$$[(\Phi_1 \circ \Phi_2)']^T J (\Phi_1 \circ \Phi_2)' = [\Phi_1' \Phi_2']^T J \Phi_1' \Phi_2' = \Phi_2'^T \Phi_1'^T J \Phi_1' \Phi_2' = \Phi_2'^T J \Phi_2' = J.$$

Thus the composition of any pair of symplectic maps is a symplectic map. The determinant of a symplectic map is ± 1 , hence these maps are always invertible, and the inverse of a symplectic map is symplectic since $\Phi'^T J \Phi' = J$ implies $J = \Phi'^{-T} J \Phi'^{-1}$. Thus the symplectic maps form a group under composition.